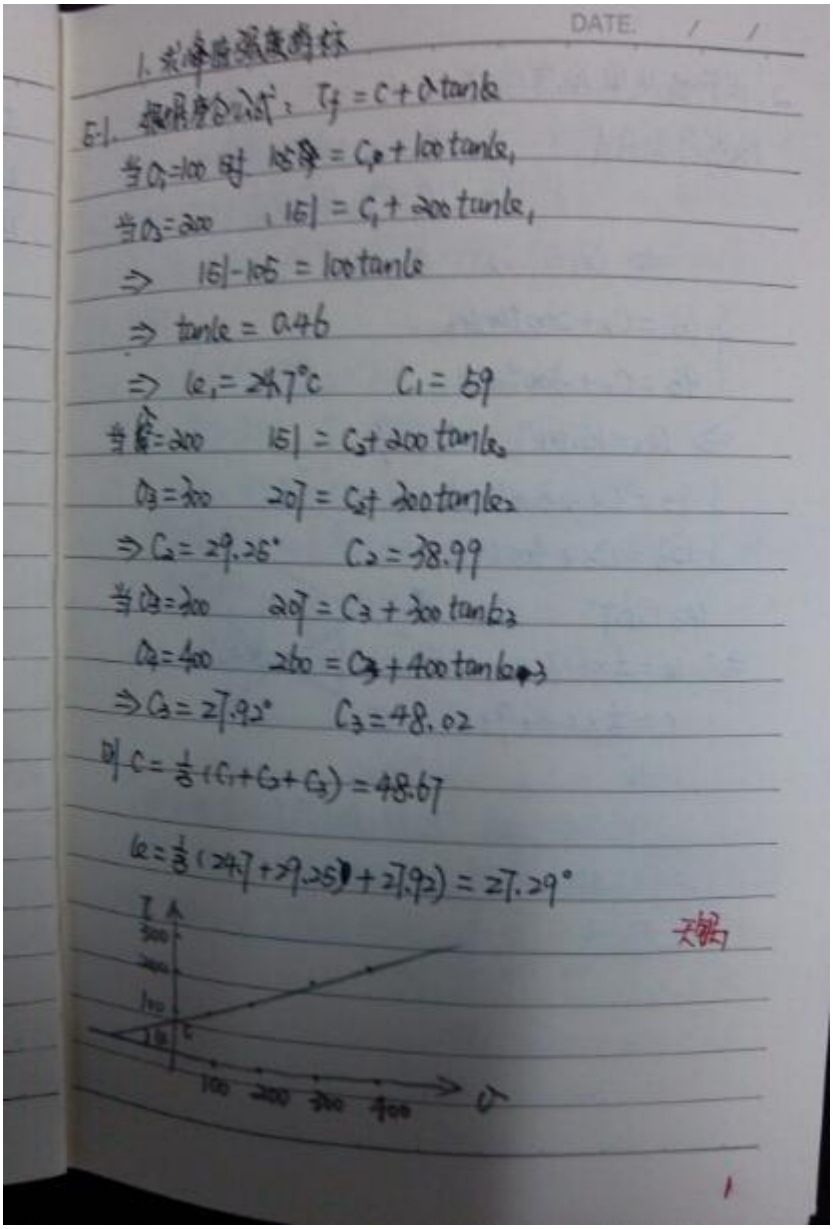

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1. 求冷却速度的方法
$t_f = C + \Delta t \tan\theta$
$\tan\theta = \frac{t_f - C}{\Delta t}$
当 $\Delta t_1 = 100$ 时 $t_{f1} = C_1 + 100\tan\theta_1$
当 $\Delta t_2 = 200$ 时 $t_{f2} = C_2 + 200\tan\theta_2$
$\Rightarrow t_{f2} - t_{f1} = 100\tan\theta_2$
$\Rightarrow \tan\theta_2 = 0.46$
$\Rightarrow \theta_2 = 24.7°$ $C_1 = 59$
当 $\Delta t_2 = 200$ $t_{f2} = C_2 + 200\tan\theta_2$
$\Delta t_3 = 300$ $t_{f3} = C_2 + 300\tan\theta_2$
$\Rightarrow C_2 = 29.26°$ $C_2 = 38.99$
当 $\Delta t_3 = 300$ $t_{f3} = C_3 + 300\tan\theta_3$
$\Delta t_4 = 400$ $t_{f4} = C_3 + 400\tan\theta_3$
$\Rightarrow \theta_3 = 27.92°$ $C_3 = 48.02$
则 $C = \frac{1}{3}(C_1 + C_2 + C_3) = 48.67$
$\theta = \frac{1}{3}(24.7 + 29.26 + 27.92) = 27.29°$
I A
300
200
100
0
100 200 300 400 t
大纲
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2、求终值抗剪强度指标

依据库仑公式 $\begin{cases} 34 = C_1 + 100\tan\varphi_1 \\ 66 = C_1 + 200\tan\varphi_1 \end{cases}$

$\Rightarrow \varphi_1 = 17.22^\circ \quad C_1 = 3$

$\begin{cases} 66 = C_2 + 200\tan\varphi_2 \\ 93 = C_2 + 300\tan\varphi_2 \end{cases}$

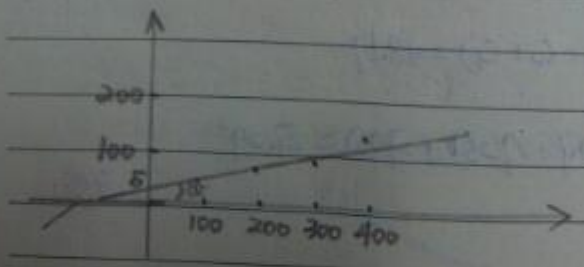
$\Rightarrow \varphi_2 = 15.64^\circ \quad C_2 = 9$

$\begin{cases} 93 = C_3 + 300\tan\varphi_3 \\ 123 = C_3 + 400\tan\varphi_3 \end{cases}$

$\varphi_3 = 16.7^\circ \quad C_3 = 3^\circ$

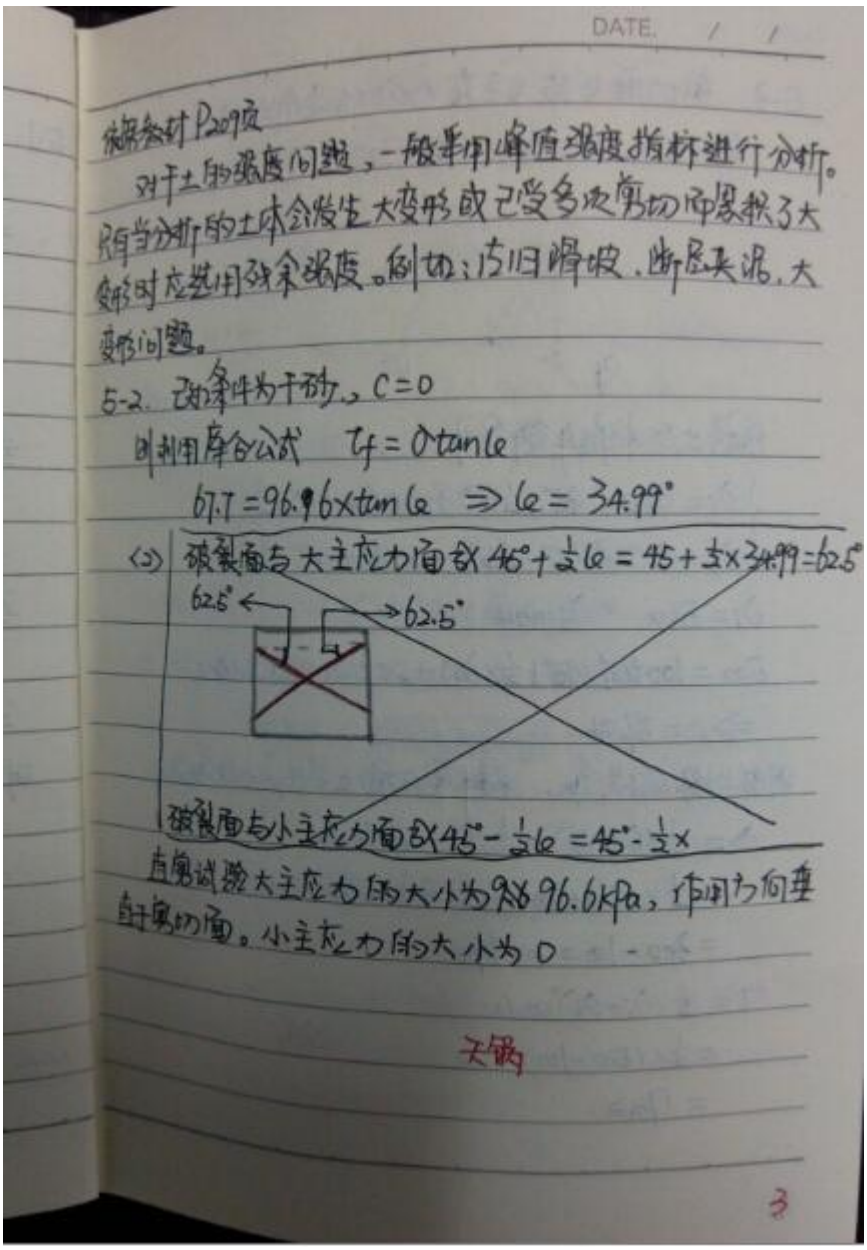
$\Rightarrow \varphi = \frac{1}{3}\times(17.22 + 15.64 + 16.7) = 16.52^\circ$

$C = \frac{1}{3}\times(3+9+3) = 5$



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依据教材P209页

对于土的强度问题，一般采用峰值强度指标进行分析。

但当分析的土体会发生大变形或已受多次剪切而累积了大变形时应采用残余强度。例如：古旧滑坡、断层夹泥、大变形问题。

5-2. 已知条件为干砂，$C=0$

则利用摩尔公式 $\tau_f = \sigma \tan\varphi$

$67.7 = 96.96 \times \tan\varphi \Rightarrow \varphi = 34.99°$

(2) 破裂面与大主应力面夹角 $45° + \frac{1}{2}\varphi = 45 + \frac{1}{2}\times 34.99 = 62.5°$

破裂面与小主应力面夹角 $45° - \frac{1}{2}\varphi = 45° - \frac{1}{2}\times$

直角试验大主应力的大小为 96.6kPa，作用方向垂直于剪切面。小主应力的大小为 0

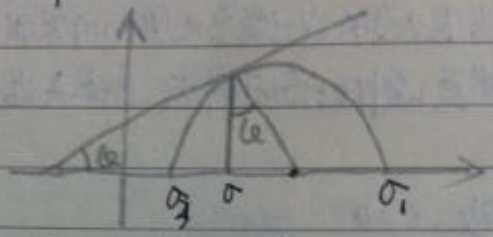
完毕

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5-3. 剪切面与最大主应力面的夹角为 $45°+\frac{1}{2}\varphi$

即 $45°+\frac{1}{2}\varphi=60$　$\varphi=30°$



依据土的极限平衡条件：

$$\begin{cases}\sigma_1=\sigma_3\tan^2(45°+\frac{1}{2}\varphi)+2c\tan(45°+\frac{\varphi}{2})\\ \sigma_3=\sigma_1\tan^2(45°-\frac{1}{2}\varphi)-2c\tan(45°-\frac{\varphi}{2})\end{cases}$$

$\sigma_1=500$　$\sigma_3=100$

$500=100\tan^2(45°+\frac{1}{2}\times30)+2c\tan(45°+\frac{1}{2}\times30)$

$\Rightarrow c=57.74$

利用上图几何关系，求解剪切面上法向应力和剪应力

$\sigma=\frac{1}{2}(\sigma_1+\sigma_3)-\frac{1}{2}(\sigma_1-\sigma_3)\sin\varphi$

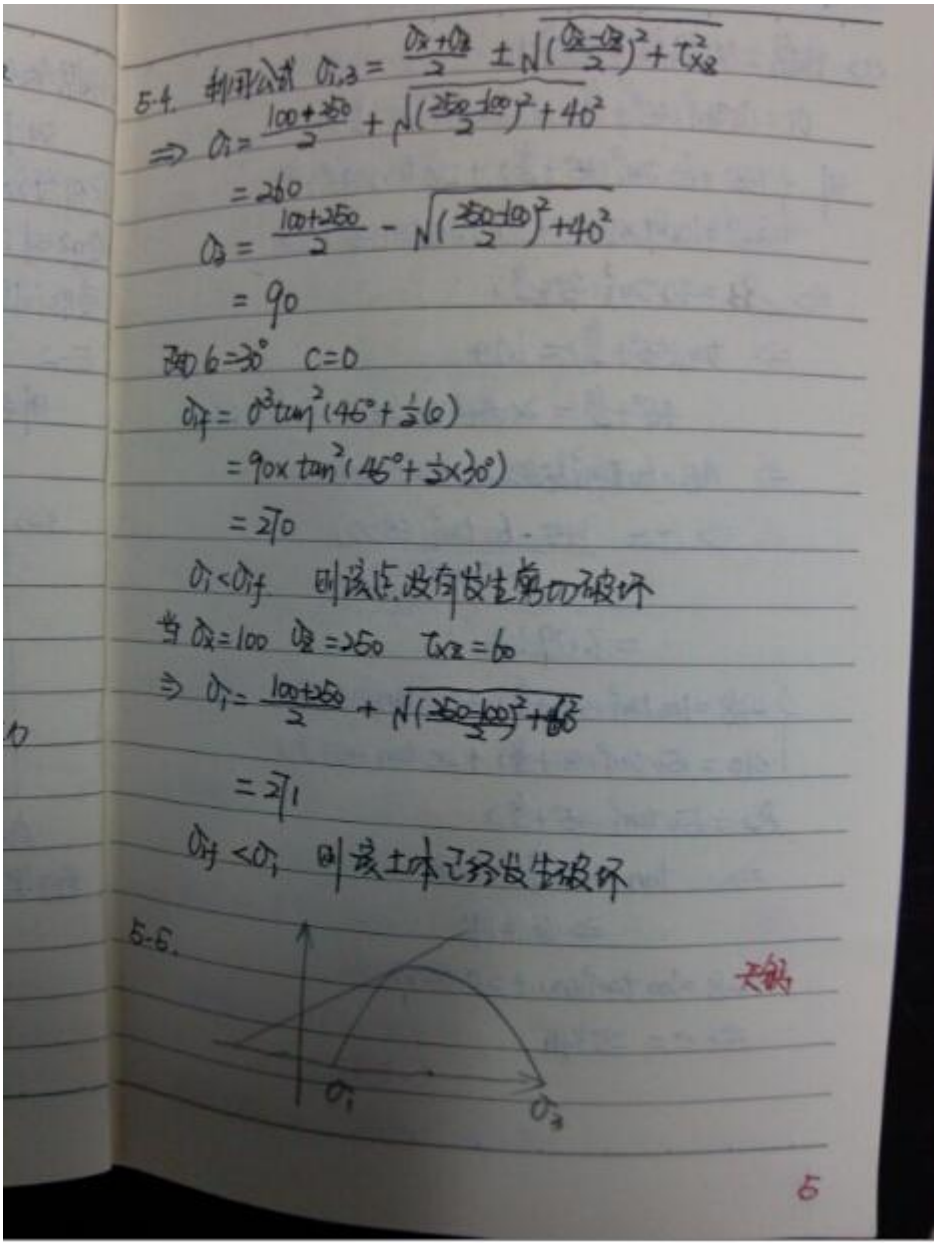
$=\frac{1}{2}\times(500+100)-\frac{1}{2}(500-100)\times\sin30$

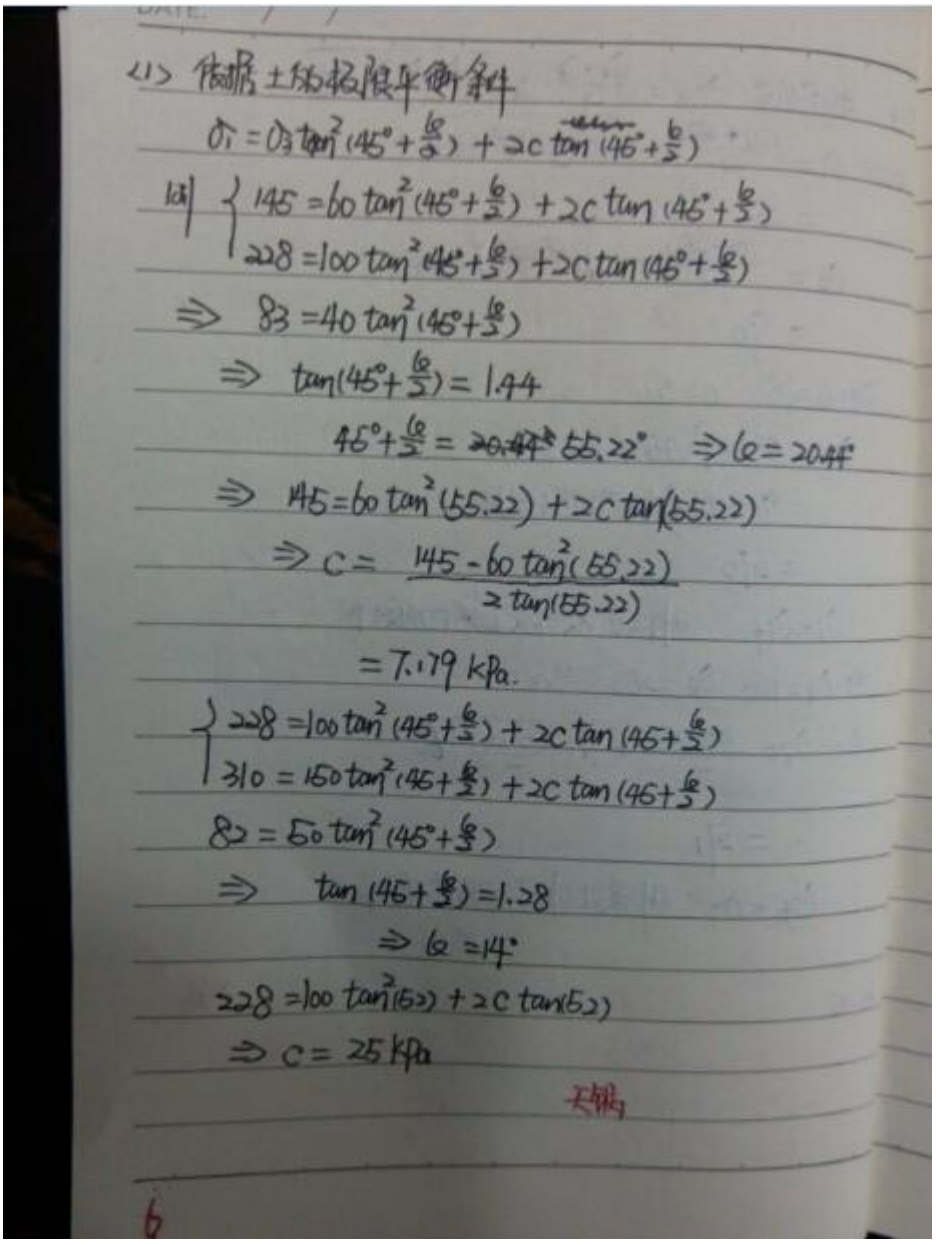
$=300-100=200\,kPa$

$\tau=\frac{1}{2}(\sigma_1-\sigma_3)\cos\varphi$

$=\frac{1}{2}\times(500-100)\cos30$　王钢

$=173.2$


5-4. 利用公式 $\sigma_{1,3}=\frac{\sigma_x+\sigma_z}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_z}{2})^2+\tau_{xz}^2}$
$\Rightarrow \sigma_1=\frac{100+350}{2}+\sqrt{(\frac{350-100}{2})^2+40^2}$
$=260$
$\sigma_3=\frac{100+350}{2}-\sqrt{(\frac{350-100}{2})^2+40^2}$
$=90$
因 $\theta=30°$ $c=0$
$\sigma_f=\sigma_3\tan^2(45°+\frac{1}{2}\theta)$
$=90\times\tan^2(45°+\frac{1}{2}\times30°)$
$=270$
$\sigma_1<\sigma_f$ 则该点没有发生剪切破坏
当 $\sigma_x=100$ $\sigma_z=250$ $\tau_{xz}=60$
$\Rightarrow \sigma_1=\frac{100+350}{2}+\sqrt{(\frac{350-100}{2})^2+60^2}$
$=271$
$\sigma_f<\sigma_1$ 则该土体已经发生破坏
5-6.
正确
σ_1
σ_3
5



(1) 饱和土极限平衡条件

$\sigma_1 = \sigma_3 \tan^2(45° + \frac{\varphi}{2}) + 2c \tan(45° + \frac{\varphi}{2})$

由 $\begin{cases} 145 = 60 \tan^2(45° + \frac{\varphi}{2}) + 2c \tan(45° + \frac{\varphi}{2}) \\ 228 = 100 \tan^2(45° + \frac{\varphi}{2}) + 2c \tan(45° + \frac{\varphi}{2}) \end{cases}$

$\Rightarrow 83 = 40 \tan^2(45° + \frac{\varphi}{2})$

$\Rightarrow \tan(45° + \frac{\varphi}{2}) = 1.44$

$45° + \frac{\varphi}{2} = 55.22°$　$\Rightarrow \varphi = 20.44°$

$\Rightarrow 145 = 60 \tan^2(55.22) + 2c \tan(55.22)$

$\Rightarrow c = \frac{145 - 60 \tan^2(55.22)}{2 \tan(55.22)}$

$= 7.179\ kPa.$

由 $\begin{cases} 228 = 100 \tan^2(45° + \frac{\varphi}{2}) + 2c \tan(45° + \frac{\varphi}{2}) \\ 310 = 150 \tan^2(45° + \frac{\varphi}{2}) + 2c \tan(45° + \frac{\varphi}{2}) \end{cases}$

$82 = 50 \tan^2(45° + \frac{\varphi}{2})$

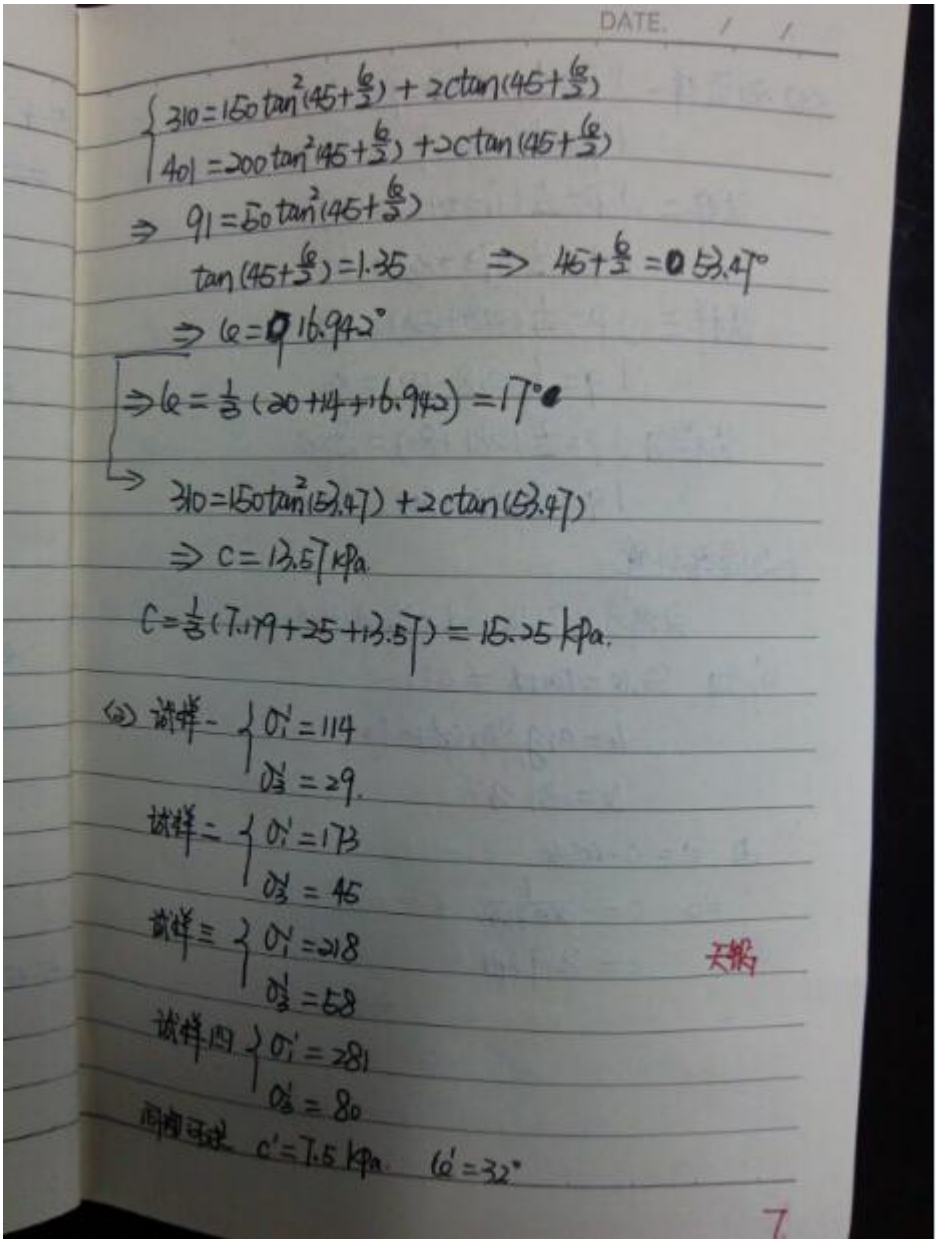
$\Rightarrow \tan(45° + \frac{\varphi}{2}) = 1.28$

$\Rightarrow \varphi = 14°$

$228 = 100 \tan^2(52) + 2c \tan(52)$

$\Rightarrow c = 25\ kPa$

6



$$\begin{cases} 310 = 150\tan^2(45+\frac{\varphi}{2}) + 2c\tan(45+\frac{\varphi}{2}) \\ 401 = 200\tan^2(45+\frac{\varphi}{2}) + 2c\tan(45+\frac{\varphi}{2}) \end{cases}$$

$$\Rightarrow 91 = 50\tan^2(45+\frac{\varphi}{2})$$

$$\tan(45+\frac{\varphi}{2}) = 1.35 \quad \Rightarrow 45+\frac{\varphi}{2} = 53.47^\circ$$

$$\Rightarrow \varphi = 16.942^\circ$$

$$\Rightarrow \varphi = \frac{1}{3}(20+14+16.942) = 17^\circ$$

$$310 = 150\tan^2(53.47) + 2c\tan(53.47)$$

$$\Rightarrow c = 13.57\ \text{kPa}$$

$$C = \frac{1}{3}(7.179+25+13.57) = 15.25\ \text{kPa}$$

(2) 试样一 $\begin{cases} \sigma_1' = 114 \\ \sigma_3' = 29 \end{cases}$

试样二 $\begin{cases} \sigma_1' = 173 \\ \sigma_3' = 45 \end{cases}$

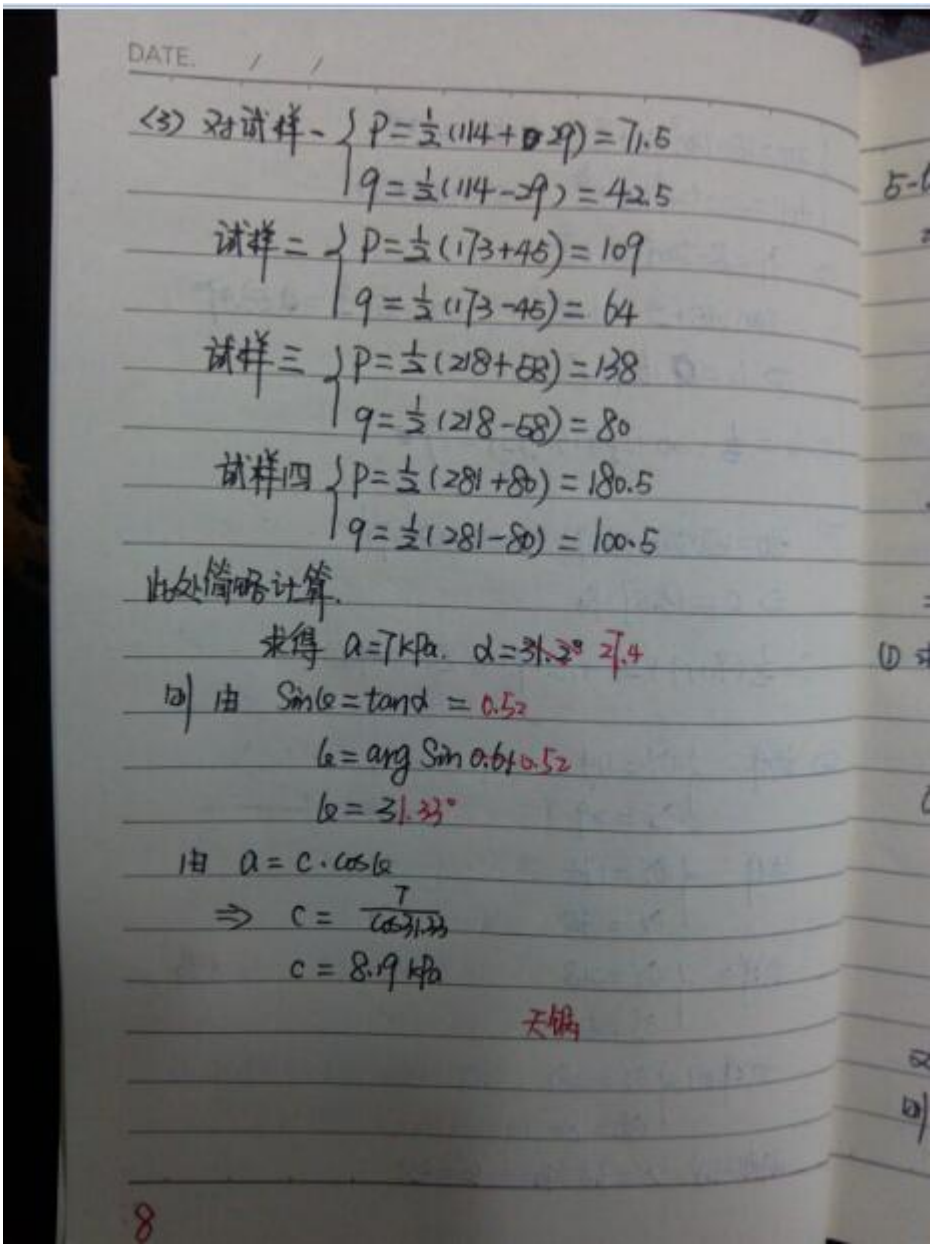
试样三 $\begin{cases} \sigma_1' = 218 \\ \sigma_3' = 58 \end{cases}$

试样四 $\begin{cases} \sigma_1' = 281 \\ \sigma_3' = 80 \end{cases}$

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两面可求 $c' = 7.5\ \text{kPa}$　$\varphi' = 32^\circ$

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(3) 对试样一 $\begin{cases} P=\frac{1}{2}(114+29)=71.5 \\ q=\frac{1}{2}(114-29)=42.5 \end{cases}$

试样二 $\begin{cases} P=\frac{1}{2}(173+45)=109 \\ q=\frac{1}{2}(173-45)=64 \end{cases}$

试样三 $\begin{cases} P=\frac{1}{2}(218+58)=138 \\ q=\frac{1}{2}(218-58)=80 \end{cases}$

试样四 $\begin{cases} P=\frac{1}{2}(281+80)=180.5 \\ q=\frac{1}{2}(281-80)=100.5 \end{cases}$

此处简略计算.

求得 $a=7kPa$. $\alpha=31.3°$ 2.4

b) 由 $\sin\varphi=\tan\alpha=0.52$

$\varphi=\arg\sin 0.64\ 0.52$

$\varphi=31.33°$

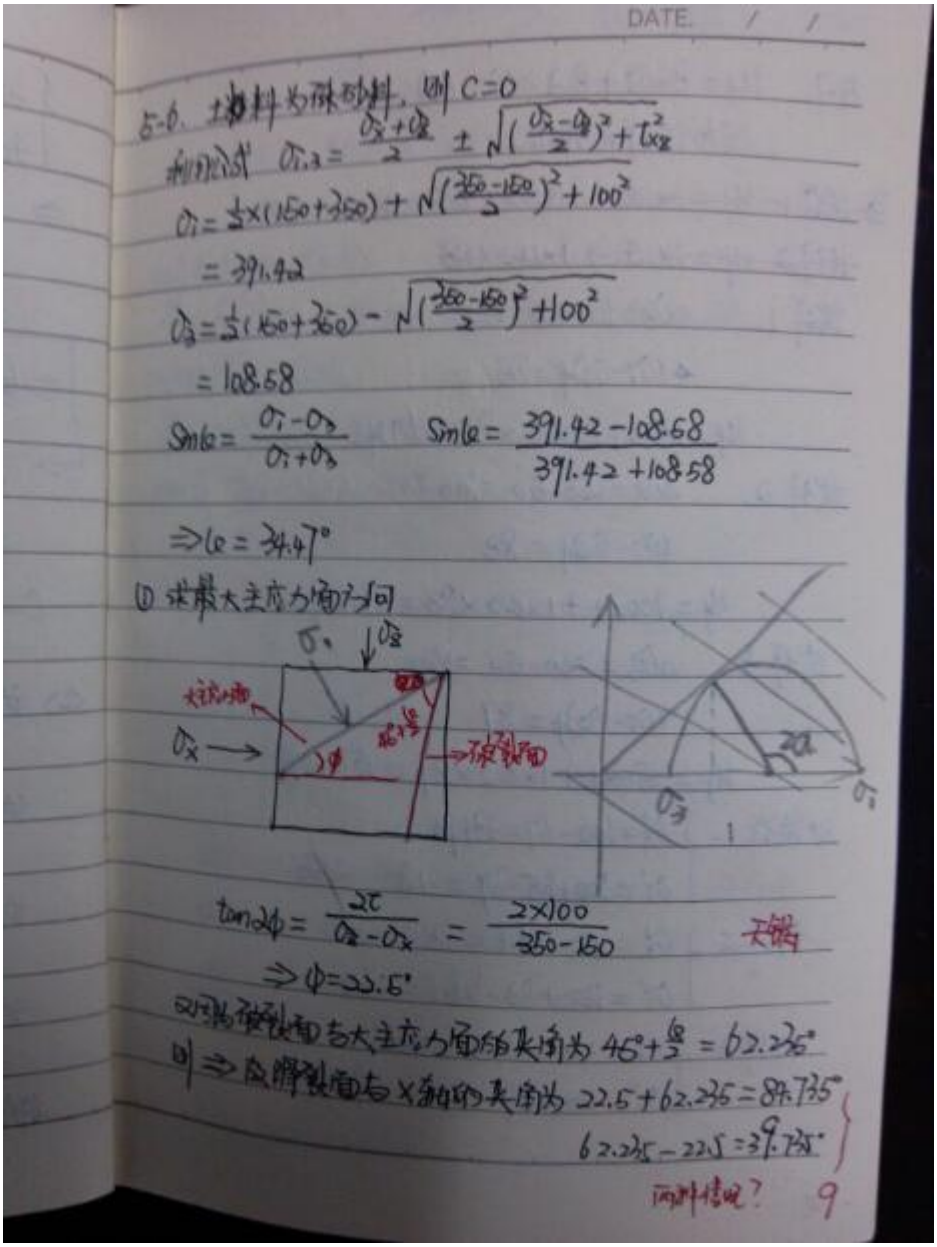
由 $a=c\cdot\cos\varphi$

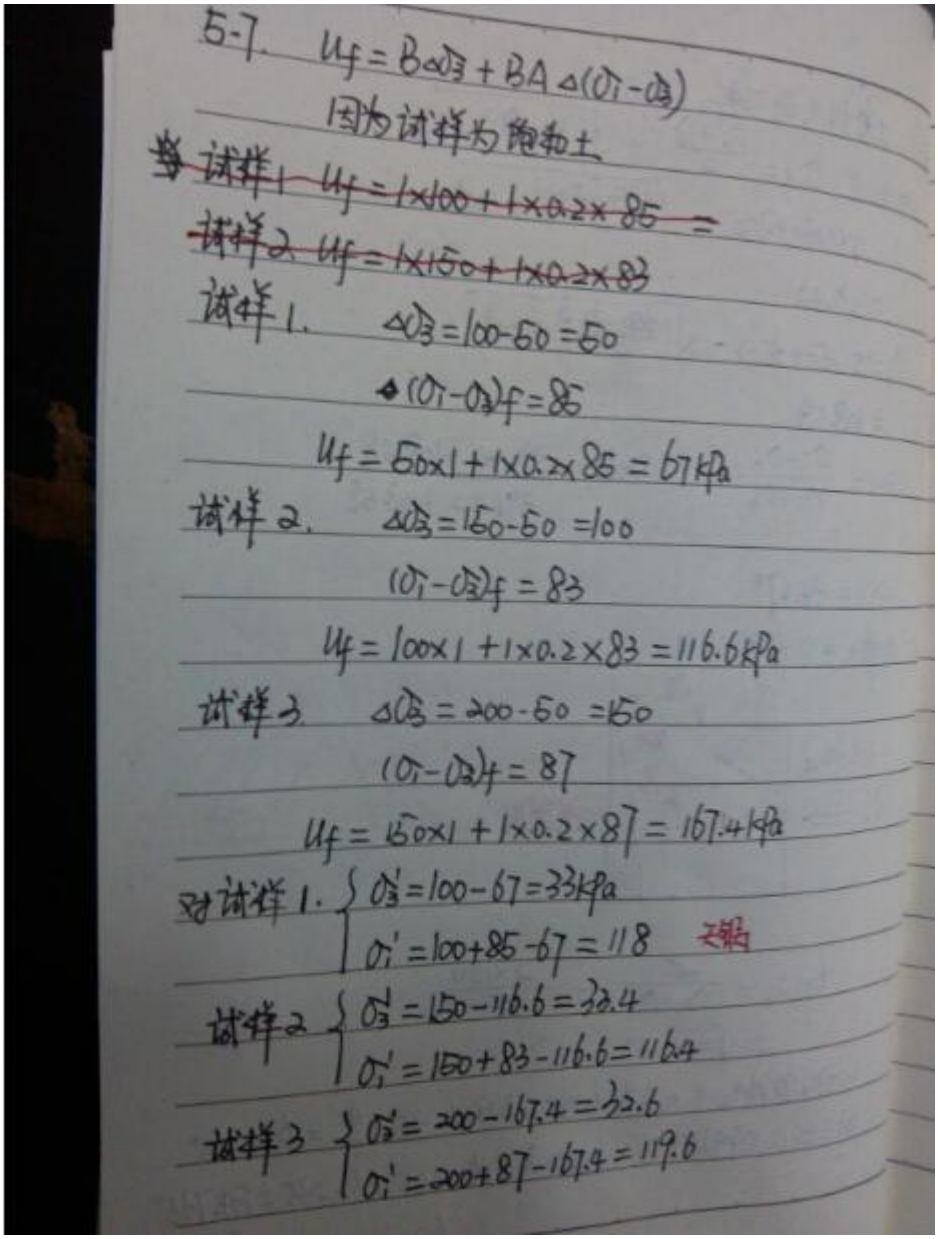
$\Rightarrow c=\frac{7}{\cos 31.33}$

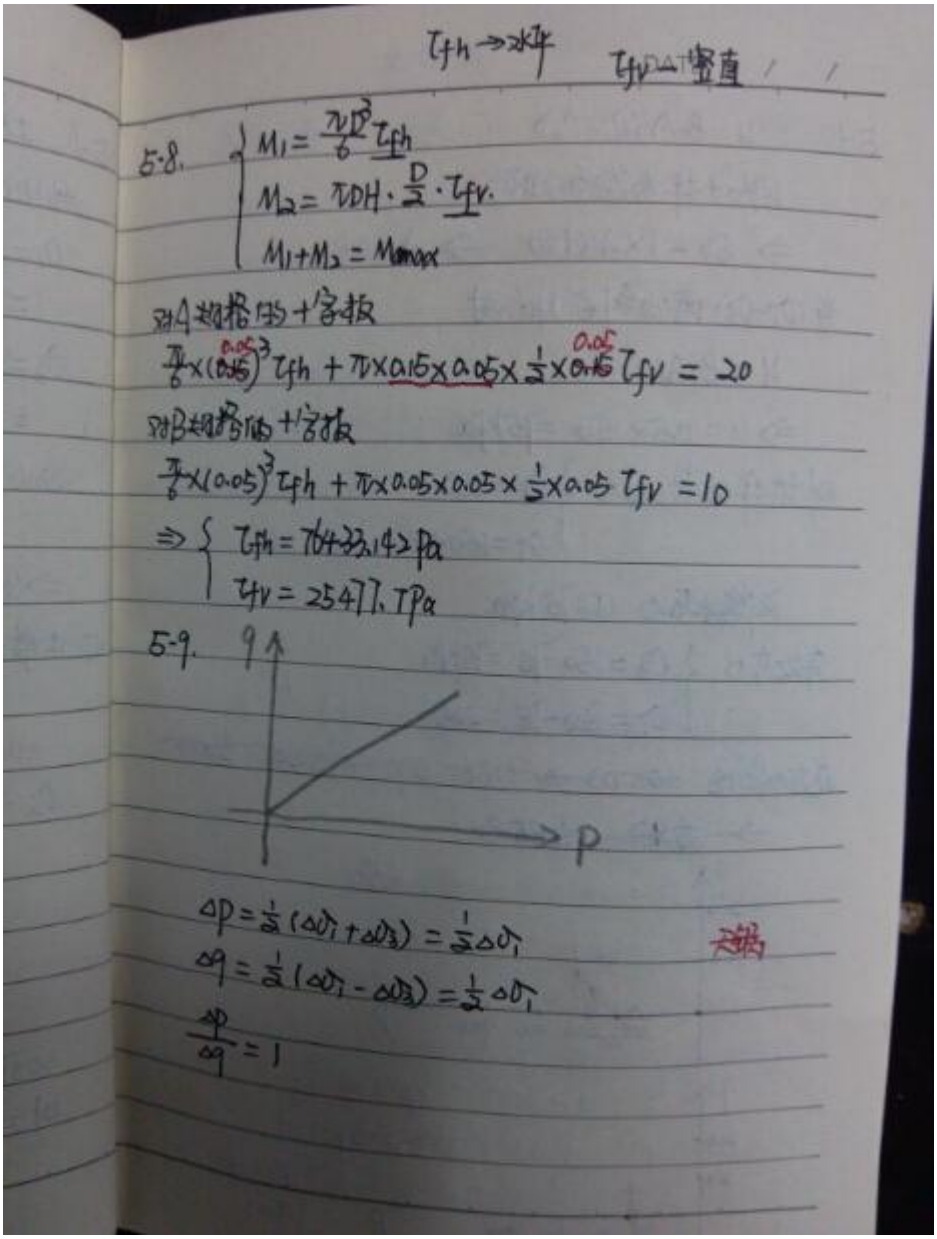
$c=8.19kPa$

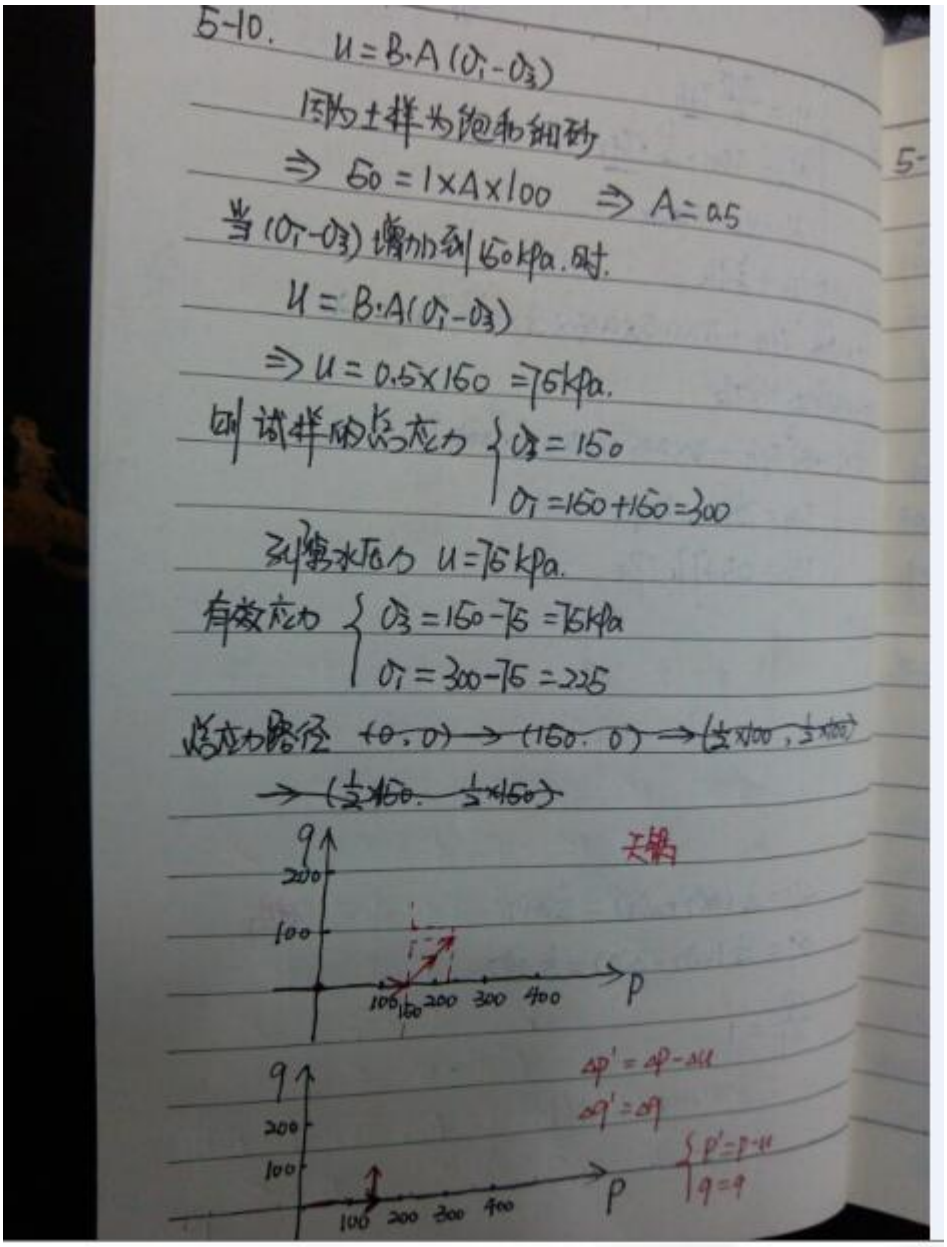
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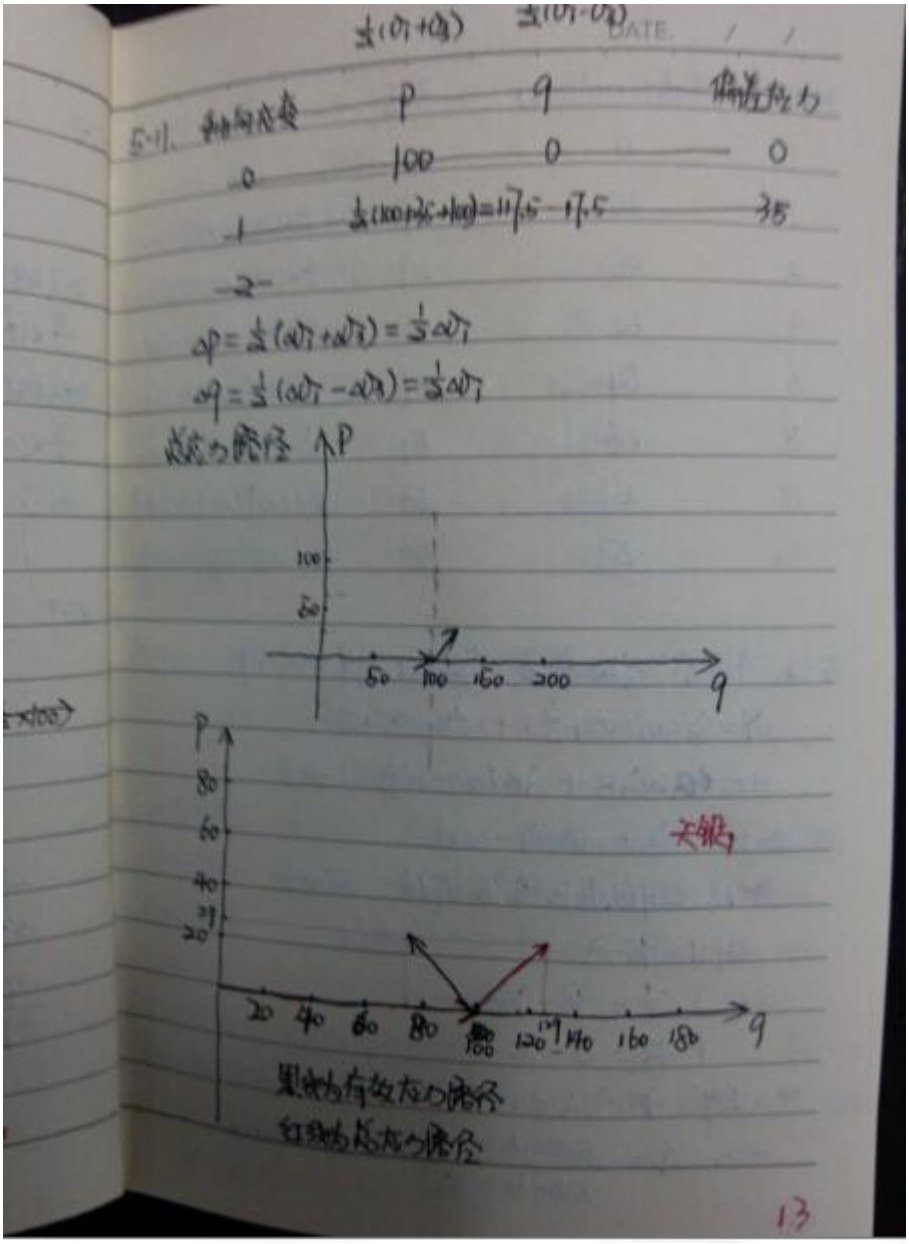
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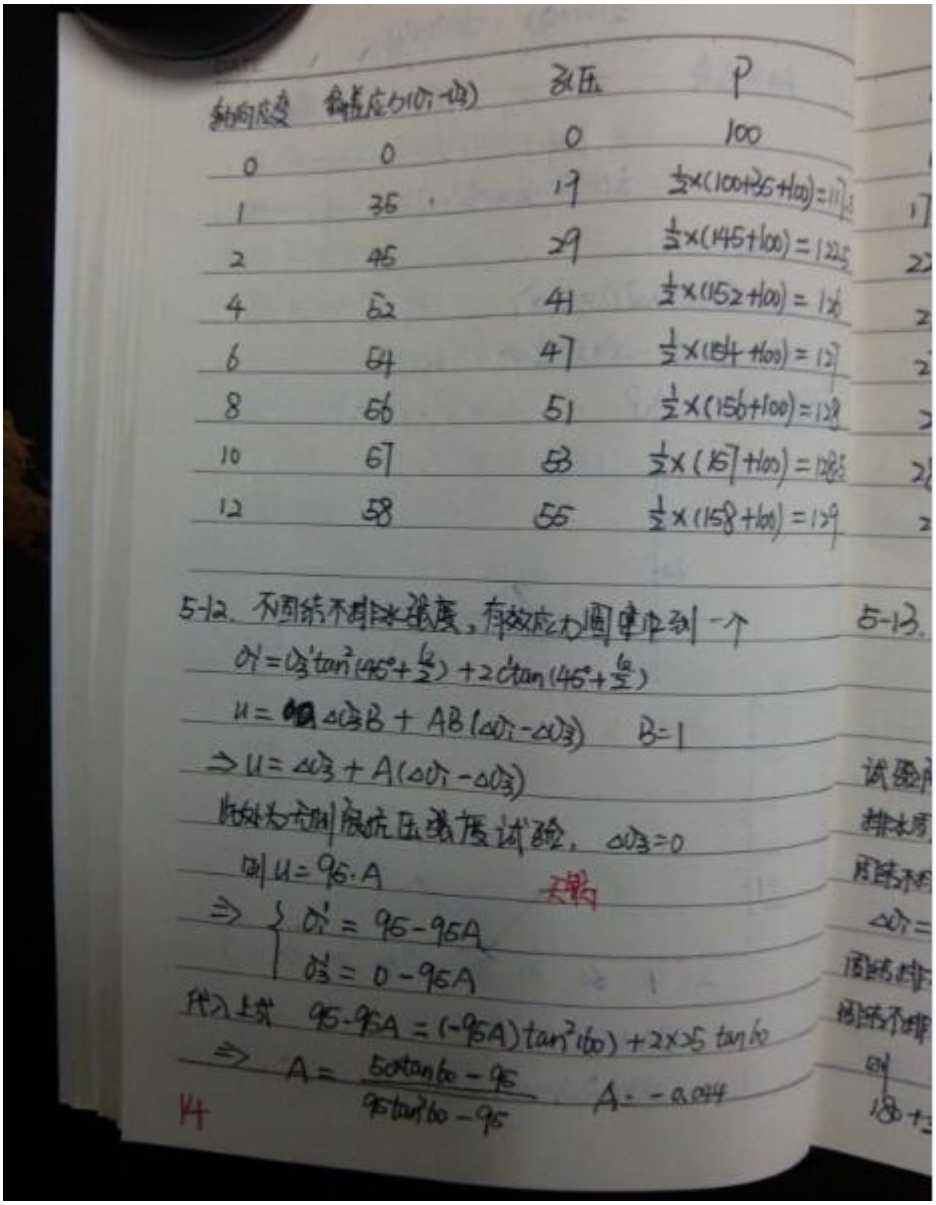

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5-6. 土料为理想材料，则 C=0
利用公式 $\sigma_{1,3}=\frac{\sigma_x+\sigma_z}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_z}{2})^2+\tau_{xz}^2}$
$\sigma_1=\frac{1}{2}\times(150+350)+\sqrt{(\frac{350-150}{2})^2+100^2}$
$=391.42$
$\sigma_3=\frac{1}{2}(150+350)-\sqrt{(\frac{350-150}{2})^2+100^2}$
$=108.58$
$\sin\varphi=\frac{\sigma_1-\sigma_3}{\sigma_1+\sigma_3}$　$\sin\varphi=\frac{391.42-108.58}{391.42+108.58}$
$\Rightarrow\varphi=34.7°$
① 求最大主应力面方向
σ_1
σ_2
大主应力面
σ_x
破裂面
σ_3
σ_1
$\tan2\psi=\frac{2\tau}{\sigma_z-\sigma_x}=\frac{2\times100}{350-150}$
$\Rightarrow\psi=22.5°$
即破裂面与大主应力面的夹角为 $45°+\frac{\varphi}{2}=62.235°$
即⇒破裂面与x轴的夹角为 $22.5+62.235=84.735°$
$62.235-22.5=39.735°$
两种情况？


5-7. $u_f = B\Delta\sigma_3 + BA\Delta(\sigma_1-\sigma_3)$
因为试样为饱和土
~~试样1 $u_f = 1\times100 + 1\times0.2\times85$ =~~
~~试样2 $u_f = 1\times150 + 1\times0.2\times83$~~
试样1. $\Delta\sigma_3 = 100-50 = 50$
$(\sigma_1-\sigma_3)_f = 85$
$u_f = 50\times1 + 1\times0.2\times85 = 67kPa$
试样2. $\Delta\sigma_3 = 150-50 = 100$
$(\sigma_1-\sigma_3)_f = 83$
$u_f = 100\times1 + 1\times0.2\times83 = 116.6kPa$
试样3 $\Delta\sigma_3 = 200-50 = 150$
$(\sigma_1-\sigma_3)_f = 87$
$u_f = 150\times1 + 1\times0.2\times87 = 167.4kPa$
对试样1. $\sigma_3' = 100-67 = 33kPa$
$\sigma_1' = 100+85-67 = 118$ 无侧
试样2 $\sigma_3' = 150-116.6 = 33.4$
$\sigma_1' = 150+83-116.6 = 116.4$
试样3 $\sigma_3' = 200-167.4 = 32.6$
$\sigma_1' = 200+87-167.4 = 119.6$


τfh→水平
τfv→竖直
5-8.
$\begin{cases} M_1 = \frac{\pi D^3}{6}\tau_{fh} \\ M_2 = \pi DH \cdot \frac{D}{2} \cdot \tau_{fv} \\ M_1 + M_2 = M_{max} \end{cases}$
对A规格的+字板
$\frac{\pi}{6}\times(0.05)^3\tau_{fh} + \pi\times0.15\times0.05\times\frac{1}{2}\times0.05\,\tau_{fv} = 20$
对B规格的+字板
$\frac{\pi}{6}\times(0.05)^3\tau_{fh} + \pi\times0.05\times0.05\times\frac{1}{2}\times0.05\,\tau_{fv} = 10$
$\Rightarrow \begin{cases} \tau_{fh} = 33.142\,Pa \\ \tau_{fv} = 2547.1\,Pa \end{cases}$
5-9.
q
p
$\Delta p = \frac{1}{2}(\Delta\sigma_1 + \Delta\sigma_3) = \frac{1}{2}\Delta\sigma_1$
$\Delta q = \frac{1}{2}(\Delta\sigma_1 - \Delta\sigma_3) = \frac{1}{2}\Delta\sigma_1$
$\frac{\Delta p}{\Delta q} = 1$


5-10.
$u = B \cdot A(\sigma_1 - \sigma_3)$
饱和土样为饱和细砂
$\Rightarrow 60 = 1 \times A \times 100 \Rightarrow A = 0.5$
当$(\sigma_1-\sigma_3)$增加到150kPa.时.
$u = B \cdot A(\sigma_1 - \sigma_3)$
$\Rightarrow u = 0.5 \times 150 = 75kPa.$
试样的总应力 $\sigma_3 = 150$
$\sigma_1 = 150 + 150 = 300$
孔隙水压力 $u = 75kPa.$
有效应力 $\sigma_3 = 150 - 75 = 75kPa$
$\sigma_1 = 300 - 75 = 225$
应力路径 (0,0)→(150,0)→(½×100, ½×100)
→(½×150, ½×150)
q
200
100
100 150 200 300 400
p
失败
q
200
100
100 200 300 400
p
$\Delta p' = \Delta p - \Delta u$
$\Delta q' = \Delta q$
$p' = p - u$
$q = q$


P
q
100
80
60
40
37
20
20
40
60
80
100
120
140
160
180
13



轴向应变	偏差应力($\sigma_1-\sigma_3$)	孔压	P
0	0	0	100
1	35	17	$\frac{1}{2}\times(100+135+100)=117.5$
2	45	29	$\frac{1}{2}\times(145+100)=122.5$
4	52	41	$\frac{1}{2}\times(152+100)=126$
6	54	47	$\frac{1}{2}\times(154+100)=127$
8	56	51	$\frac{1}{2}\times(156+100)=128$
10	57	53	$\frac{1}{2}\times(157+100)=128.5$
12	58	55	$\frac{1}{2}\times(158+100)=129$

5-12. 不固结不排水强度，有效应力圆半径到一个

$\sigma_1'=\sigma_3'\tan^2(45°+\frac{\varphi}{2})+2c'\tan(45°+\frac{\varphi}{2})$

$u=\Delta\sigma_3 B+AB(\Delta\sigma_1-\Delta\sigma_3)$　$B=1$

$\Rightarrow u=\Delta\sigma_3+A(\Delta\sigma_1-\Delta\sigma_3)$

饱和为无侧限抗压强度试验，$\Delta\sigma_3=0$

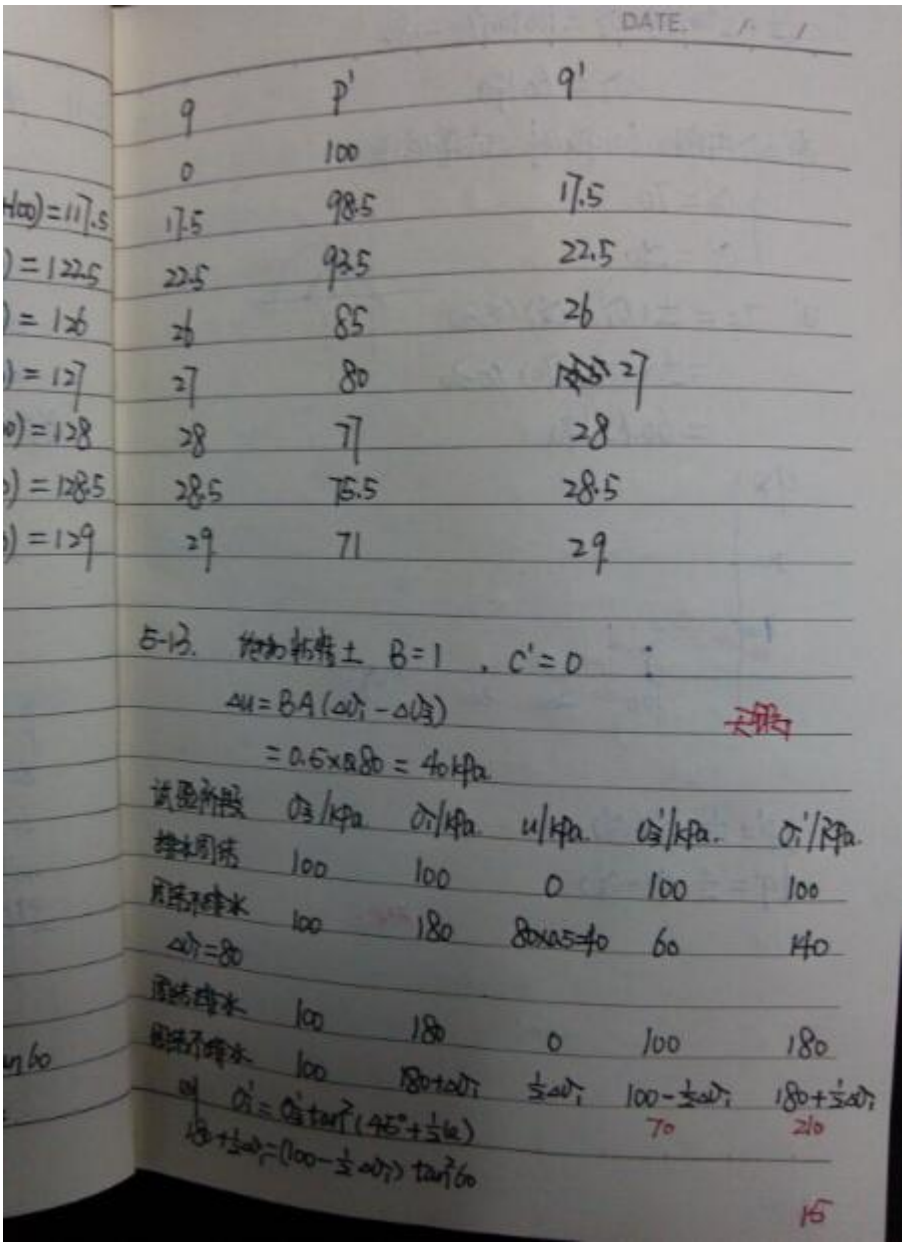
则 $u=95\cdot A$

$\Rightarrow\begin{cases}\sigma_1'=95-95A\\ \sigma_3'=0-95A\end{cases}$

代入上式 $95-95A=(-95A)\tan^2(60)+2\times5\tan60$

$\Rightarrow A=\frac{5\tan60-95}{95\tan^260-95}$　$A=-0.044$

5-13.



q	p'	q'
0	100	
17.5	98.5	17.5
22.5	93.5	22.5
26	85	26
27	80	27
28	77	28
28.5	75.5	28.5
29	71	29

5-13. 饱和粘性土 $B=1$，$c'=0$：

$\Delta u = B\Delta(\sigma_1 - \sigma_3)$

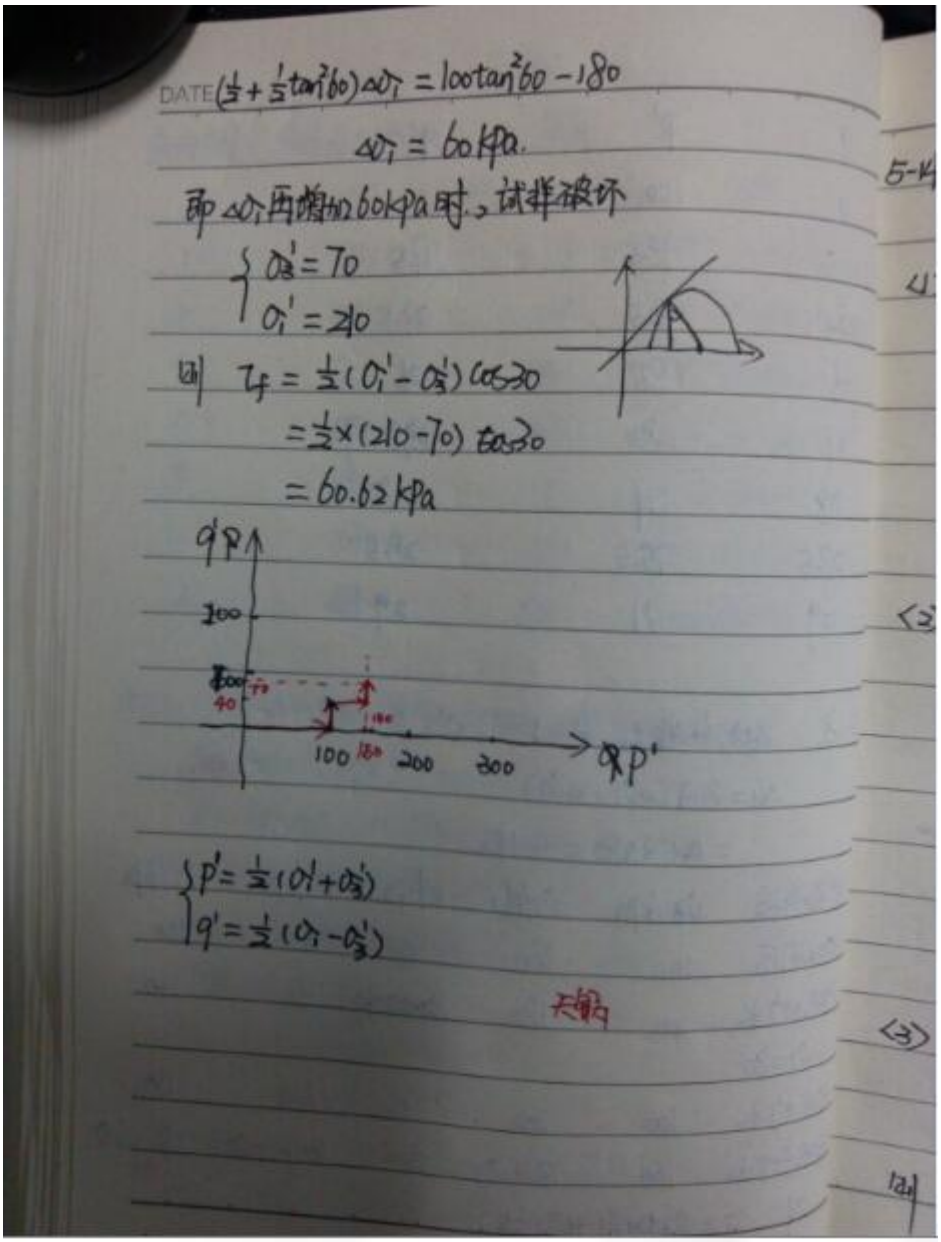
$= 0.6\times 80 = 40\ kPa$

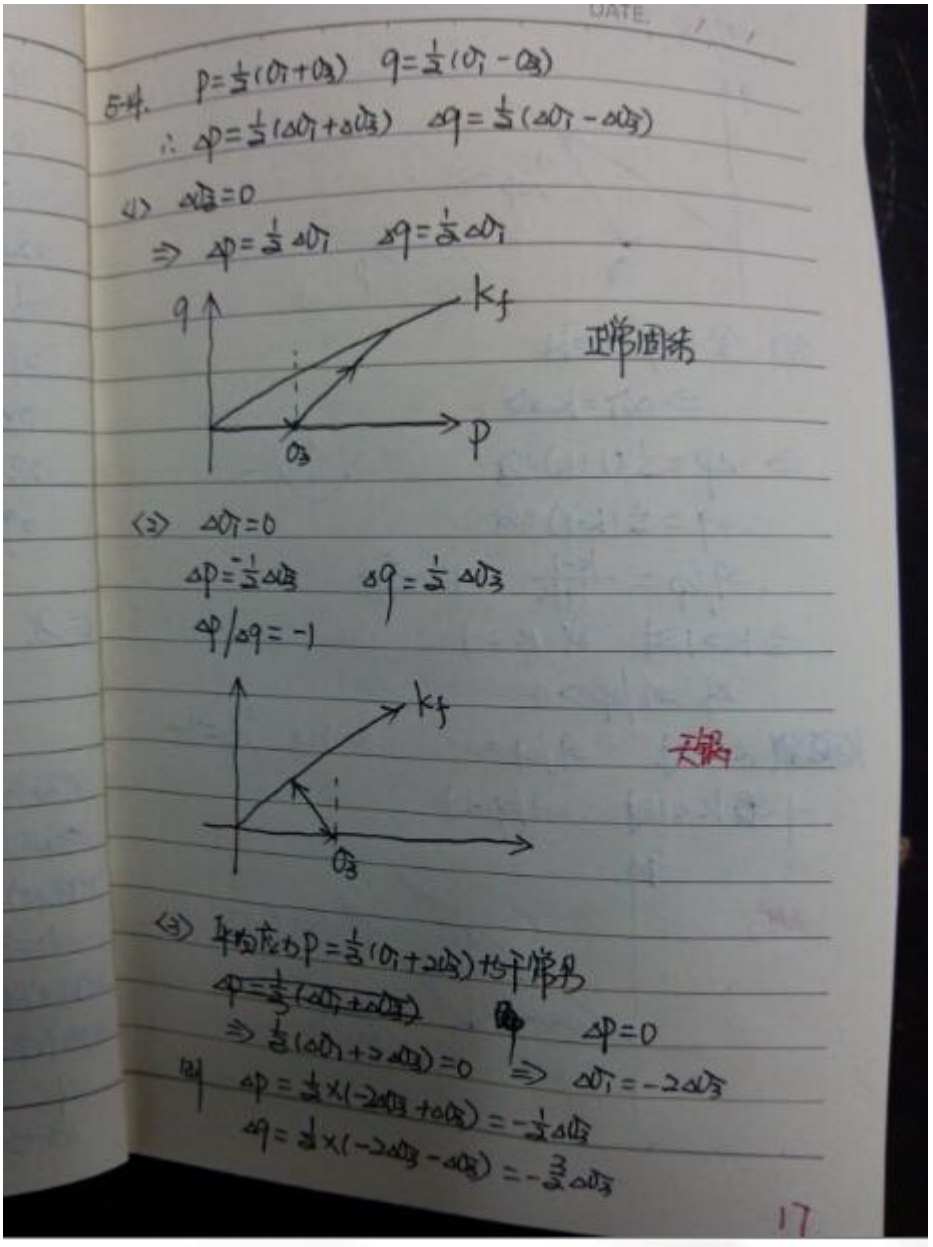
试验阶段	σ_3/kPa	σ_1/kPa	u/kPa	σ_3'/kPa	σ_1'/kPa
固结围压	100	100	0	100	100
不排水剪	100	180	80×0.5=40	60	140
$\Delta\sigma_1=80$					
固结围压	100	180	0	100	180
不排水剪	100	$180+\Delta\sigma_1$	$\frac{1}{2}\Delta\sigma_1$	$100-\frac{1}{2}\Delta\sigma_1$	$180+\frac{1}{2}\Delta\sigma_1$
				70	210

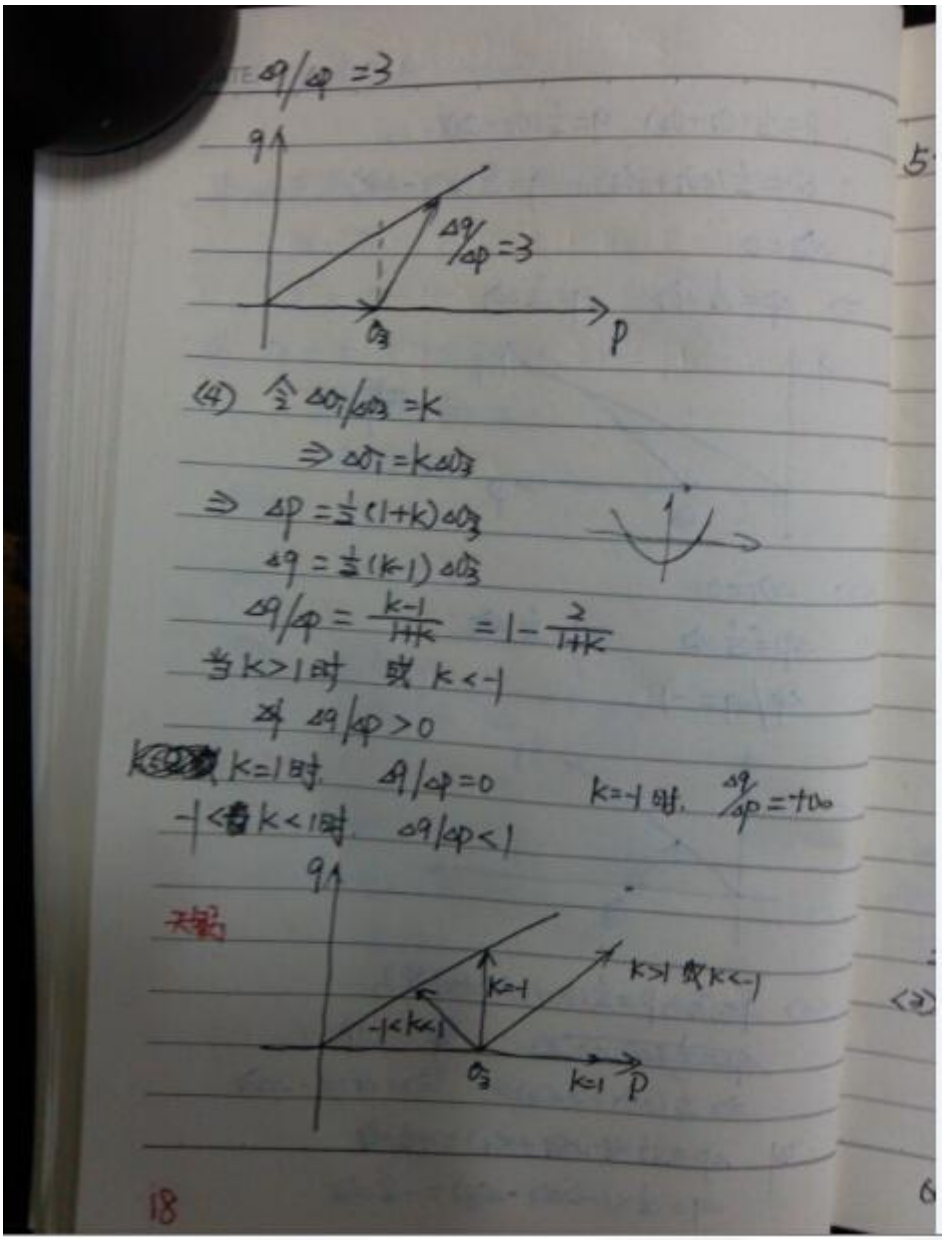
由 $\sigma_1' = \sigma_3'\tan^2(45°+\frac{1}{2}\varphi)$

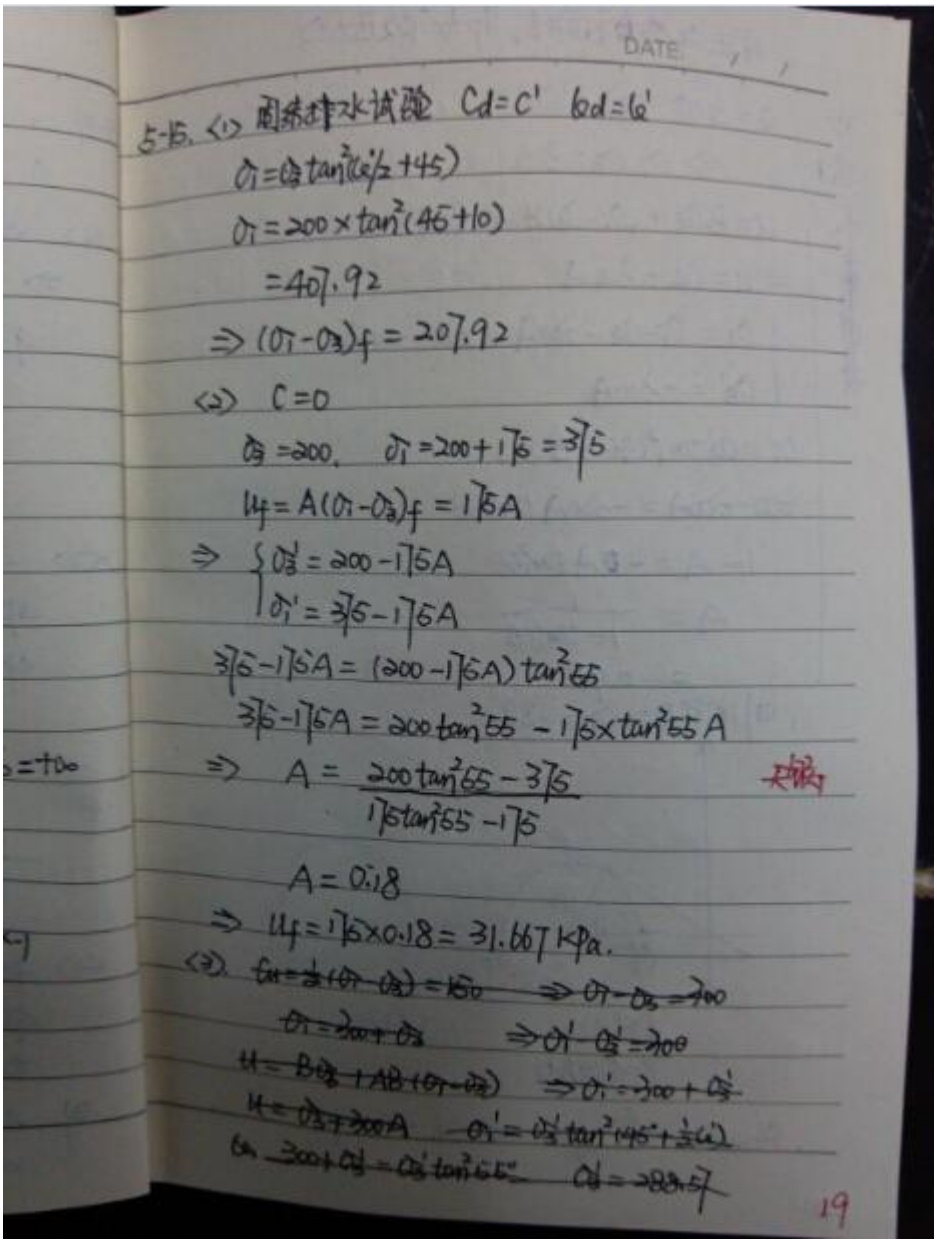
$180+\frac{1}{2}\Delta\sigma_1 = (100-\frac{1}{2}\Delta\sigma_1)\tan^2 60$

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$(\frac{1}{3}+\frac{1}{3}\tan^2 60)\Delta\sigma_1 = 100\tan^2 60 - 180$
$\Delta\sigma_1 = 60kPa$
即 $\Delta\sigma_1$ 再增加 60kPa 时，试样破坏
$\sigma_3' = 70$
$\sigma_1' = 210$
$\tau_f = \frac{1}{2}(\sigma_1' - \sigma_3')\cos 30$
$= \frac{1}{2}\times(210-70)\cos 30$
$= 60.62kPa$
q/p
200
100
40
100
180
200
300
p'
$p' = \frac{1}{2}(\sigma_1' + \sigma_3')$
$q' = \frac{1}{2}(\sigma_1 - \sigma_3')$


5-4. $P=\frac{1}{2}(Q_1+Q_2)$ $Q=\frac{1}{2}(Q_1-Q_2)$
$\therefore \Delta P=\frac{1}{2}(\Delta Q_1+\Delta Q_2)$ $\Delta Q=\frac{1}{2}(\Delta Q_1-\Delta Q_2)$
(1) $\Delta Q_2=0$
$\Rightarrow \Delta P=\frac{1}{2}\Delta Q_1$ $\Delta Q=\frac{1}{2}\Delta Q_1$
Q
K_f
正常商品
Q_3
P
(2) $\Delta Q_1=0$
$\Delta P=-\frac{1}{2}\Delta Q_2$ $\Delta Q=\frac{1}{2}\Delta Q_2$
$\Delta P/\Delta Q=-1$
K_f
Q_3
(3) $\Delta P=0$
$\Rightarrow \frac{1}{2}(\Delta Q_1+2\Delta Q_2)=0$
$\Rightarrow \Delta Q_1=-2\Delta Q_2$
$\Delta P=\frac{1}{2}\times(-2\Delta Q_2+\Delta Q_2)=-\frac{1}{2}\Delta Q_2$
$\Delta Q=\frac{1}{2}\times(-2\Delta Q_2-\Delta Q_2)=-\frac{3}{2}\Delta Q_2$
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ΔQ/ΔP =3
Q
ΔQ/ΔP =3
P
(4) 令 ΔQ1/ΔQ2 =K
⇒ ΔQ1 =KΔQ2
⇒ ΔP =½(1+K)ΔQ2
ΔQ =½(K-1)ΔQ2
ΔQ/ΔP = (K-1)/(1+K) =1- 2/(1+K)
当K>1时 或 K<-1
有 ΔQ/ΔP >0
K=1时 ΔQ/ΔP =0
K=-1时 ΔQ/ΔP =+∞
-1<K<1时 ΔQ/ΔP <1
Q
K=1
-1<K<1
K>1 或 K<-1
O
K=1
P



5-15. (1) 固结不排水试验 $C_d=C'$ $\varphi_d=\varphi'$

$\sigma_1=\sigma_3\tan^2(\varphi/2+45)$

$\sigma_1=200\times\tan^2(45+10)$

$=407.92$

$\Rightarrow(\sigma_1-\sigma_3)_f=207.92$

(2) $C=0$

$\sigma_3=200$, $\sigma_1=200+175=375$

$u_f=A(\sigma_1-\sigma_3)_f=175A$

$\Rightarrow\begin{cases}\sigma_3'=200-175A\\ \sigma_1'=375-175A\end{cases}$

$375-175A=(200-175A)\tan^2 55$

$375-175A=200\tan^2 55-175\times\tan^2 55\,A$

$\Rightarrow A=\dfrac{200\tan^2 55-375}{175\tan^2 55-175}$

$A=0.18$

$\Rightarrow u_f=175\times0.18=31.667\,kPa.$

(3) ~~$f_f=\frac{1}{2}(\sigma_1-\sigma_3)=150\Rightarrow\sigma_1-\sigma_3=300$~~

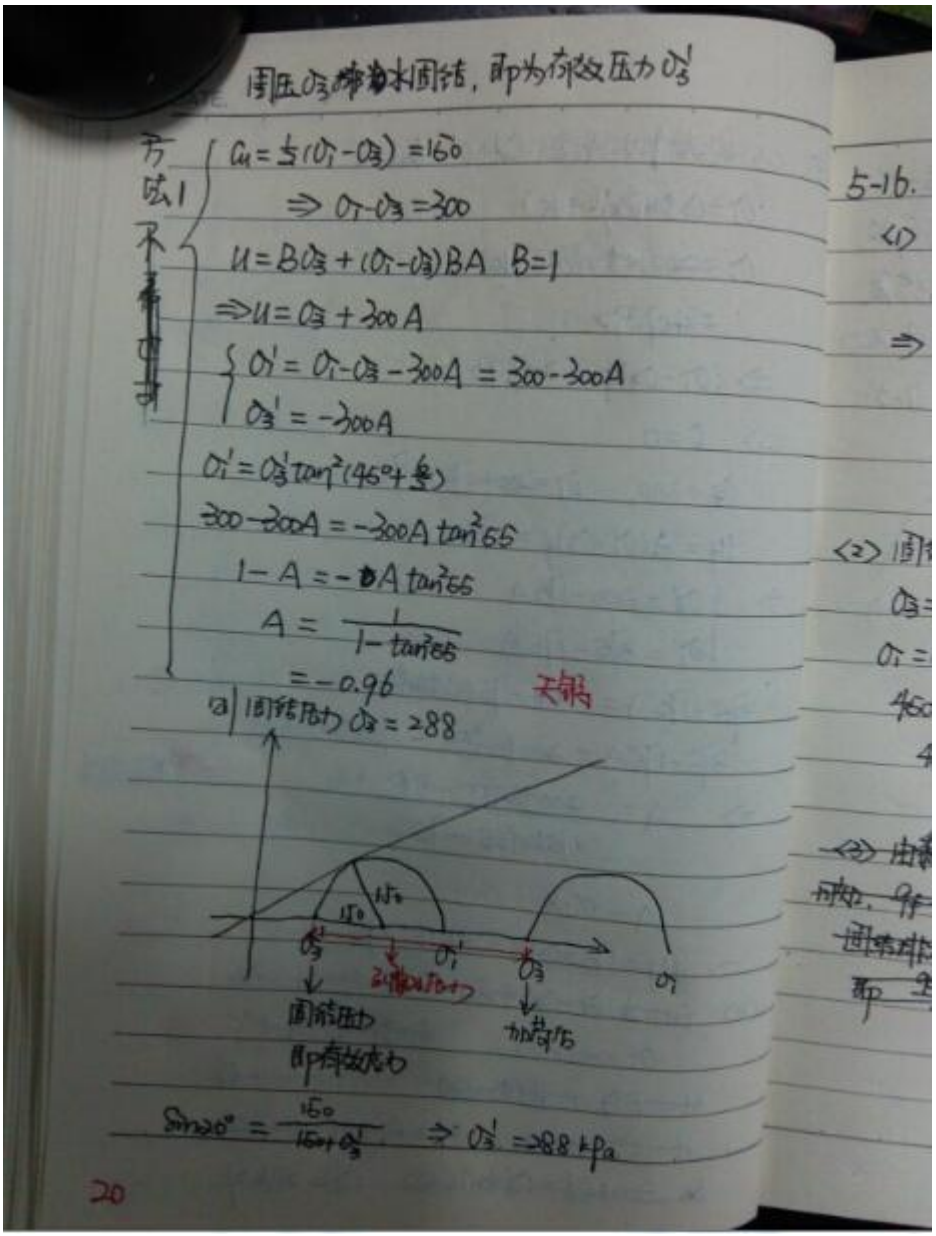
~~$\sigma_1=300+\sigma_3\Rightarrow\sigma_1'-\sigma_3'=300$~~

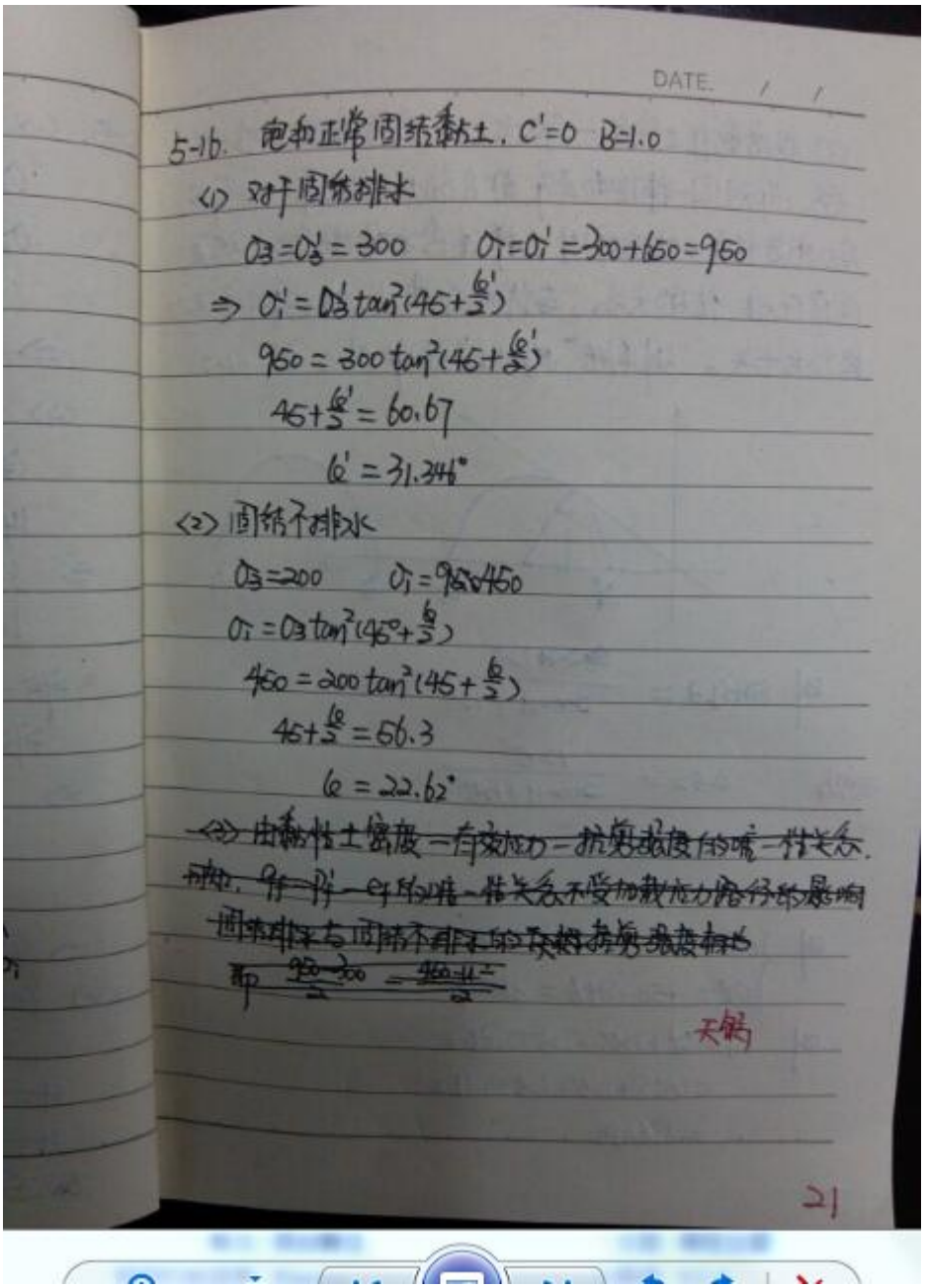
~~$u=B\sigma_3+AB(\sigma_1-\sigma_3)\Rightarrow\sigma_1'=300+\sigma_3'$~~

~~$u=\sigma_3+300A\quad\sigma_1'=\sigma_3'\tan^2(45^\circ+\frac{\varphi}{2})$~~

~~$\sigma_3+300+\sigma_3=\sigma_3'\tan^2 55^\circ\quad\sigma_3=283.57$~~

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围压σ3缓慢加到固结，即为有效压力σ3'
方法1
不排水剪
Cu = 1/2(σ1-σ3) = 150
⇒ σ1-σ3 = 300
u = Bσ3 + (σ1-σ3)BA　B=1
⇒u = σ3 + 300A
σ1' = σ1-σ3-300A = 300-300A
σ3' = -300A
σ1' = σ3' tan²(45°+φ/2)
300-300A = -300A tan²65
1-A = -3A tan²65
A = 1/(1-tan²65)
= -0.96
无解
(2) 固结压力 σ3 = 288
150
150
σ3'
σ1'
σ3
σ1
固结压力
即有效应力
加荷后
sin20° = 150/(150+σ3')
⇒ σ3' = 288 kPa
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5-16. 饱和正常固结黏土，$C'=0$　$\beta=1.0$

(1) 对于固结排水

$\sigma_3=\sigma_3'=300$　　$\sigma_1=\sigma_1'=300+650=950$

$\Rightarrow \sigma_1'=\sigma_3\tan^2(45+\frac{\phi'}{2})$

$950=300\tan^2(45+\frac{\phi'}{2})$

$45+\frac{\phi'}{2}=60.67$

$\phi'=31.34°$

(2) 固结不排水

$\sigma_3=200$　　$\sigma_1=950-450$

$\sigma_1=\sigma_3\tan^2(45°+\frac{\phi}{2})$

$450=200\tan^2(45+\frac{\phi}{2})$

$45+\frac{\phi}{2}=56.3$

$\phi=22.62°$

~~(3) 由饱和土密度=有效应力=抗剪强度的唯一性关系。~~

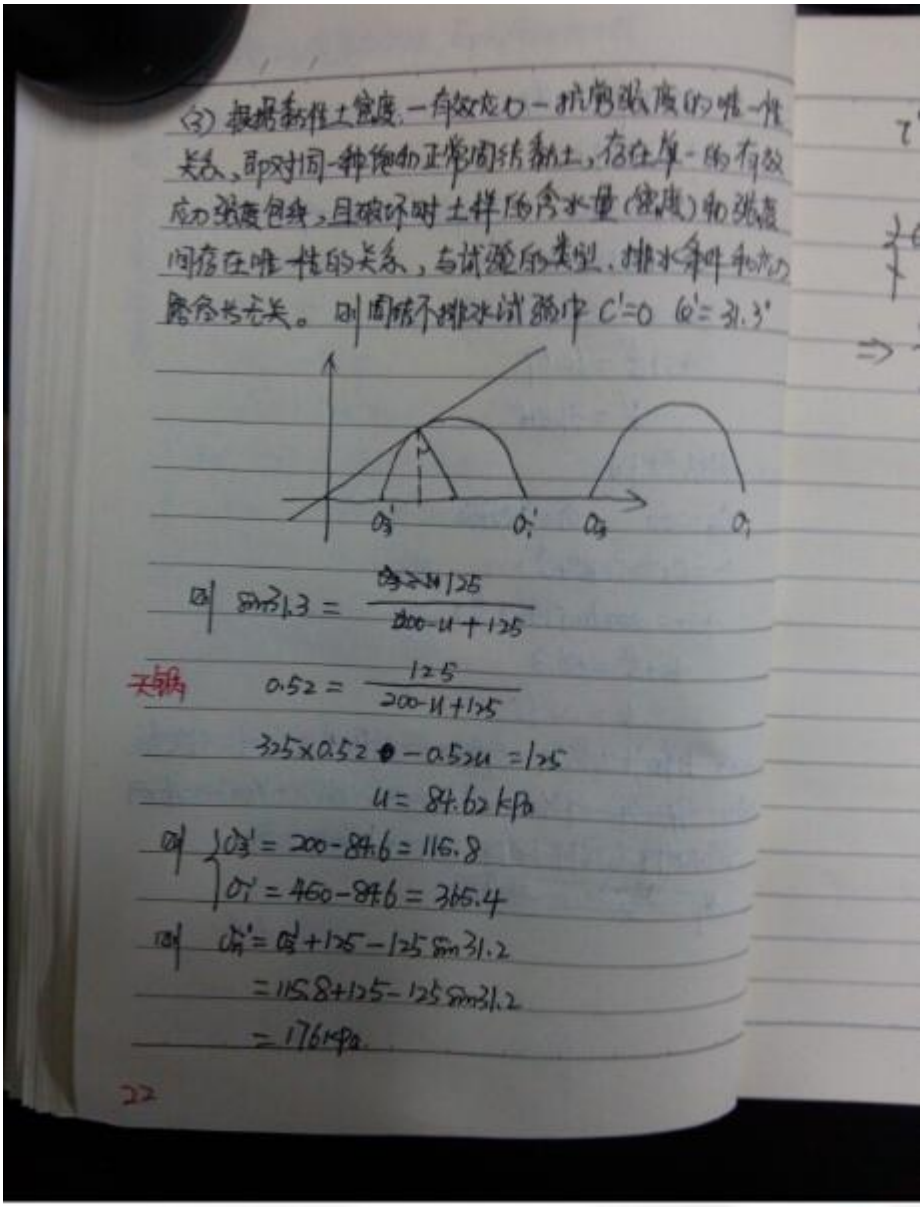
~~而此，有一个ϕ'的唯一指关系不受加载应力路径的影响~~

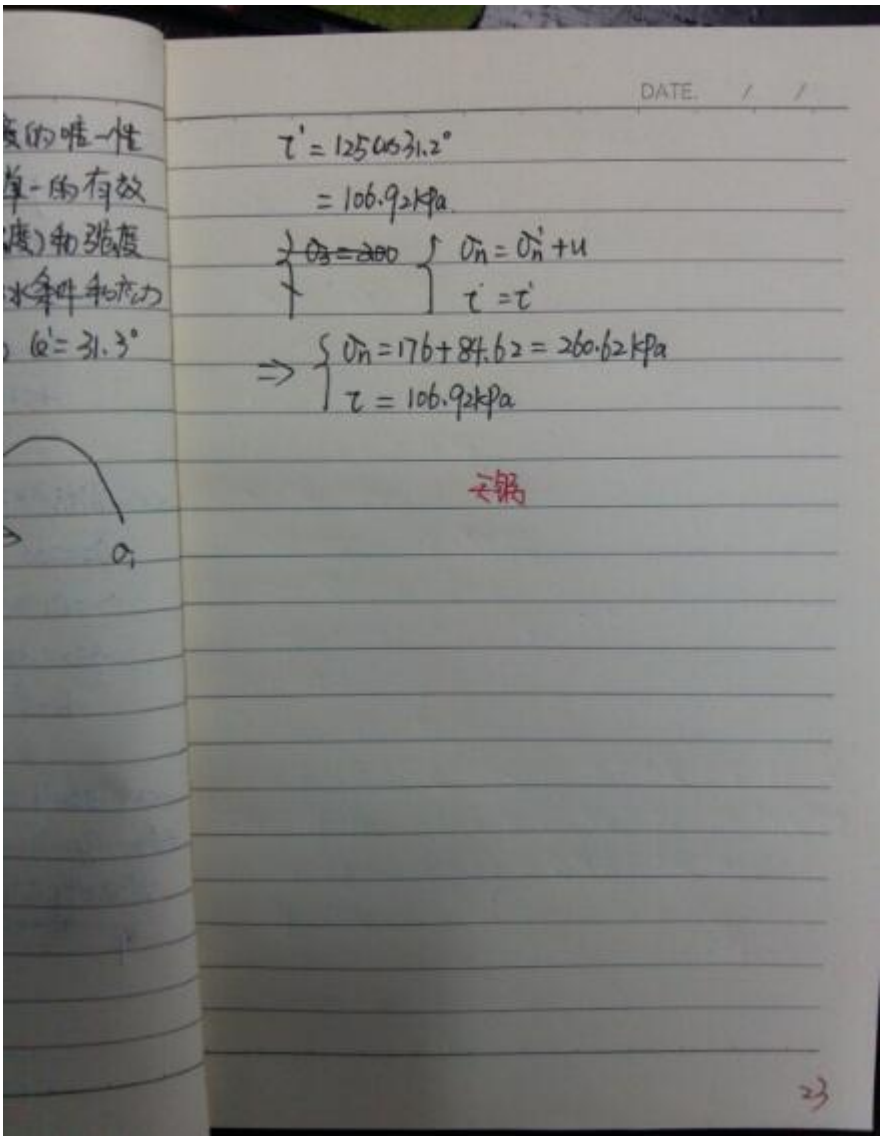
~~固结排水与固结不排水的实验的剪切强度相比~~

~~即 $\frac{950-300}{2}=\frac{950+?}{2}$~~

天解

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③ 根据黏性土宽度、有效应力、抗剪强度的唯一性关系，即对同一种饱和正常固结黏土，存在唯一的有效应力强度包线，且破坏时土样的含水量（密度）和强度间存在唯一性的关系，与试验的类型、排水条件和应力路径无关。则固结不排水试验中 c'=0 φ'=31.3°
σ_3'
σ_1'
σ_3
σ_1
则 $\sin 31.3 = \frac{\sigma_3 - u + 125}{\sigma_1 - u + 125}$
天锅
$0.52 = \frac{125}{200 - u + 125}$
$325 \times 0.52 - 0.52u = 125$
$u = 84.62\ kPa$
则 $\sigma_3' = 200 - 84.6 = 116.8$
$\sigma_1' = 460 - 84.6 = 365.4$
则 $\sigma_f' = \sigma_3' + 125 - 125\sin 31.2$
$= 115.8 + 125 - 125\sin 31.2$
$= 176\ kPa$
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DATE
τ' = 125.0tan31.2°
= 106.92kPa
σ3 = 200
σn = σn' + u
τ = τ'
⇒ σn = 176 + 84.62 = 260.62kPa
τ = 106.92kPa
σ1